

Perceptron with basis expansion

Learning in the higher-dimensional feature space:

- $w = 0$ and $b = 0$
- while some $y(w \cdot \Phi(x) + b) \leq 0$:
 - $w = w + y \Phi(x)$
 - $b = b + y$

Problem: number of features has now increased dramatically.
For MNIST, with quadratic boundary: from 784 to 308504.

The kernel trick: implement this without ever writing down a vector in the higher-dimensional space!

The kernel trick

- $w = 0$ and $b = 0$
- while some $y^{(i)}(w \cdot \Phi(x^{(i)}) + b) \leq 0$:
 - $w = w + y^{(i)} \Phi(x^{(i)})$
 - $b = b + y^{(i)}$

- ① Represent w in **dual** form: $\alpha = (\alpha_1, \dots, \alpha_n)$.

$$w = \sum_{j=1}^n \alpha_j y^{(j)} \Phi(x^{(j)})$$

- ② Compute $w \cdot \Phi(x)$ using the dual representation.

$$w \cdot \Phi(x) = \sum_{j=1}^n \alpha_j y^{(j)} (\Phi(x^{(j)}) \cdot \Phi(x))$$

- ③ Compute $\Phi(x) \cdot \Phi(z)$ without ever writing out $\Phi(x)$ or $\Phi(z)$.

Computing dot products

Suppose $x = (x_1, x_2, \dots, x_d)$ and

$$\Phi(x) = (1, \sqrt{2}x_1, \dots, \sqrt{2}x_d, x_1^2, \dots, x_d^2, \sqrt{2}x_1x_2, \dots, \sqrt{2}x_{d-1}x_d)$$

$$\begin{aligned}\Phi(x) \cdot \Phi(z) &= (1, \sqrt{2}x_1, \dots, \sqrt{2}x_d, x_1^2, \dots, x_d^2, \sqrt{2}x_1x_2, \dots, \sqrt{2}x_{d-1}x_d) \cdot \\ &\quad (1, \sqrt{2}z_1, \dots, \sqrt{2}z_d, z_1^2, \dots, z_d^2, \sqrt{2}z_1z_2, \dots, \sqrt{2}z_{d-1}z_d) \\ &= 1 + 2 \sum_i x_i z_i + \sum_i x_i^2 z_i^2 + 2 \sum_{i \neq j} x_i x_j z_i z_j \\ &= (1 + x_1 z_1 + \dots + x_d z_d)^2 = (1 + x \cdot z)^2\end{aligned}$$

For MNIST:

We are computing dot products in 308504-dimensional space.
But it takes time proportional to 784, the original dimension!

Kernel Perceptron

Learning from data $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)}) \in \mathcal{X} \times \{-1, 1\}$

Primal form:

- $w = 0$ and $b = 0$
- while there is some i with $y^{(i)}(w \cdot \Phi(x^{(i)}) + b) \leq 0$:
 - $w = w + y^{(i)} \Phi(x^{(i)})$
 - $b = b + y^{(i)}$

Dual form: $w = \sum_j \alpha_j y^{(j)} \Phi(x^{(j)})$, where $\alpha \in \mathbb{R}^n$

- $\alpha = 0$ and $b = 0$
- while some i has $y^{(i)} \left(\sum_j \alpha_j y^{(j)} \Phi(x^{(j)}) \cdot \Phi(x^{(i)}) + b \right) \leq 0$:
 - $\alpha_i = \alpha_i + 1$
 - $b = b + y^{(i)}$

To classify a new point x : $\text{sign} \left(\sum_j \alpha_j y^{(j)} \Phi(x^{(j)}) \cdot \Phi(x) + b \right)$.

Does this work with SVMs?

$$\begin{aligned} \text{(PRIMAL)} \quad & \min_{w \in \mathbb{R}^d, b \in \mathbb{R}, \xi \in \mathbb{R}^n} \quad \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.:} \quad & y^{(i)}(w \cdot x^{(i)} + b) \geq 1 - \xi_i \quad \text{for all } i = 1, 2, \dots, n \\ & \xi \geq 0 \end{aligned}$$

$$\begin{aligned} \text{(DUAL)} \quad & \max_{\alpha \in \mathbb{R}^n} \quad \sum_{i=1}^n \alpha_i - \sum_{i,j=1}^n \alpha_i \alpha_j y^{(i)} y^{(j)} (x^{(i)} \cdot x^{(j)}) \\ \text{s.t.:} \quad & \sum_{i=1}^n \alpha_i y^{(i)} = 0 \\ & 0 \leq \alpha_i \leq C \end{aligned}$$

Solution: $w = \sum_i \alpha_i y^{(i)} x^{(i)}$.

Kernel SVM

- 1 **Basis expansion.** Mapping $x \mapsto \Phi(x)$.
- 2 **Learning.** Solve the dual problem:

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^n} \quad & \sum_{i=1}^n \alpha_i - \sum_{i,j=1}^n \alpha_i \alpha_j y^{(i)} y^{(j)} (\Phi(x^{(i)}) \cdot \Phi(x^{(j)})) \\ \text{s.t.:} \quad & \sum_{i=1}^n \alpha_i y^{(i)} = 0 \\ & 0 \leq \alpha_i \leq C \end{aligned}$$

This yields $w = \sum_i \alpha_i y^{(i)} \Phi(x^{(i)})$. Offset b also follows.

- 3 **Classification.** Given a new point x , classify as

$$\text{sign} \left(\sum_i \alpha_i y^{(i)} (\Phi(x^{(i)}) \cdot \Phi(x)) + b \right).$$